Problem

Assignment 2. Car Tracking

Gary Geunbae Lee CSED342 - Artificial Intelligence
Contact: TA Yejin Min(yeajinmin@postech.ac.kr), Sujin Woo(happysujin@postech.ac.kr)

Part A

Before moving onto the main part of the assignment (Part B), you will complete several simple review questions. Refer to the class slides before answering the questions. Write your answers to Word and save the file as "PartA_studentID.pdf".

Note: There is no partial credit in this section of the assignment.

Write all answers either as irreducible fractions or rounded to three decimal places.

A-1 Probability

Q1. Jackpot? (2pts)

A casino has a slot machine that operates in “jackpot mode” only 1 hour out of every 40. When the machine is in jackpot mode, its lights flash with a probability of 80%. Even when it is not in jackpot mode, due to occasional malfunctions the lights may still flash, with a probability of 10%. One customer reports seeing the slot machine’s lights flashing. What is the probability that the machine is actually in jackpot mode at this time?

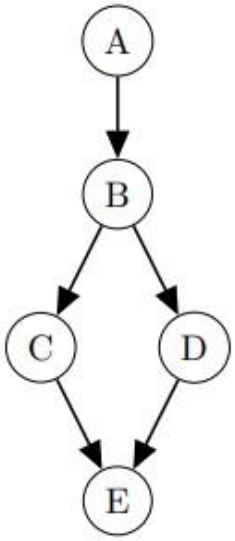
Problem

Instruction: Be sure to write out the full solution process. To receive credit, both the final answer and the solution must be entirely correct.

A-2 Bayes' Net

Assume we are given the following Bayes' net, with the associated conditional probability tables (CPTs).

Problem



A	P(A)
+a	0.5
-a	0.5

A	B	P(B|A)
+a	+b	0.2
+a	-b	0.8
-a	+b	0.5
-a	-b	0.5

B	C	P(C|B)
+b	+c	0.4
+b	-c	0.6
-b	+c	0.8
-b	-c	0.2

B	D	P(D|B)
+b	+d	0.2
+b	-d	0.8
-b	+d	0.2
-b	-d	0.8

C	D	E	P(E | C, D)
+c	+d	+e	0.4
+c	+d	-e	0.6
+c	-d	+e	0.6
+c	-d	-e	0.4
-c	+d	+e	0.2
-c	+d	-e	0.8
-c	-d	+e	0.6
-c	-d	-e	0.4

Problem

You are given a set of the following samples, but are not told whether they were collected with rejection sampling or likelihood weighting.

−a	−b	+c	+d	+e
−a	+b	+c	−d	+e
−a	−b	−c	−d	+e
−a	−b	+c	−d	+e
−a	+b	+c	+d	+e

Q2. Assuming these samples were generated from **rejection sampling**, what is the sample based estimate of P(+b | −a, +e)? (2pts)

Q3. Assuming these samples were generated from **likelihood weighting**, what is the sample-based estimate of P(+b | −a, +e)? (2pts)

Q4. Again, assume these samples were generated from likelihood weighting. However, you are not sure about the original CPT for P(E | C, D) given above being the CPT associated with the Bayes' Net:

Problem

With 50% chance, the CPT associated with the Bayes' Net is the original one. With the other 50% chance, the CPT is actually the
CPT below. Given this uncertainty, what is the sample-based estimate of P(+b | −a, +e)? (2pts)

C	D	E	P(E | C, D)
+c	+d	+e	0.8
+c	+d	-e	0.2
+c	-d	+e	0.4
+c	-d	-e	0.6
-c	+d	+e	0.2
-c	+d	-e	0.8
-c	-d	+e	0.6
-c	-d	-e	0.4

Q5. Suppose you choose to use Gibbs sampling to estimate P(B, E | +c, −d). Assume the CPTs are the same as the ones for parts (a) and (b). Currently your assignments are the following:

−a −b +c −d +e

Problem

Q5-1. Suppose the next step is to resample E. What is the probability that the new assignment to E will be +e? (2pts)

Q5-2. Instead, suppose the next step is to resample A. What is the probability that the new assignment to A will be +a? (2pts)

Q5-3. Instead, suppose the next step is to resample B. What is the probability that the new assignment to B will be +b? (2pts)